

GORAN OCOKOLJIĆ

Lead Web Engineer | Frontend Architecture | AI-Native Delivery

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PROFILE

Hands-on engineer and architect with 18+ years across high-traffic web platforms, product development, health-tech and AI-native software delivery. I unify what has grown organically into shared platforms and tooling, and stay close to implementation while doing it. Recent work: a Vue/Nuxt platform powering 20+ portals, a telehealth product used by real doctors and patients, and tr-harness, an autonomous workflow that takes GitHub issues through implementation, independent review and verified merge.

SELECTED IMPACT

20+	30% to 95%	~50%	185
portals on one shared platform	desktop Core Web Vitals green URLs	shorter idea-to-production	GitHub issues delivered autonomously by tr-harness, Jun-Aug 2026

EXPERIENCE

Senior Software Architect / Capability Manager | **WirelessMedia Group** Mar 2021 - Present

- Architected and led News Theme, a configurable Vue/Nuxt platform powering 20+ high-traffic portals with CMS-driven routing, a component factory and shared integration patterns; remained hands-on on difficult implementation and performance work.
- Drove Core Web Vitals improvement across portals: desktop green URLs from 30% to 95%, mobile from 20% to 55%.
- Shortened idea-to-production time by ~50% through stack modernization, quality gates, automated testing and CI/CD improvements.
- Set technical direction across 18-22 front-end engineers in 6 teams, including architecture alignment, mentoring, hiring, onboarding and engineering practices.
- Led AI-assisted engineering adoption and tooling across the front-end organization: agent workflows, code-review automation, reusable skills/rules and MCP-based tooling.

SELECTED AI & DEVELOPER TOOLING

tr-harness, autonomous dev-cycle harness | Personal R&D, built to develop Toprope, 2026 - ongoing

- Designed and built tr-harness, a PowerShell/Node.js orchestration harness that delivered 137 GitHub issues end-to-end between June and August 2026: read issue, implement, build/test gate, PR, five-lens independent review, fix, verified merge and issue close.
- Instrumented 172 runs and 240 review cycles with cost/quality telemetry: 2,735 review findings and 256 critical/high blockers fixed before merge; 83% run success; \$3.7k billed in total, median ~\$16 per successful run.
- Built trust-but-verify completion checks, bounded review cycles, deterministic resume, failure quarantine and a self-improving review knowledge base (130 lessons, 24 graduated into always-on rules through human-gated promotion).

AI test-generation & QA workflow

- Built a multi-agent workflow that turns requirements or legacy feature briefs into reviewed Playwright coverage: live-app analysis, discrepancy detection, manual cases, generated tests, independent proof review and controlled healing.
- Designed specifically against false greens: ambiguous requirements escalate to humans, generated tests must prove they can fail, and real product defects are not 'healed' by weakening assertions.

PureContext MCP

- Built PureContext, an open-source MCP code-intelligence server for AI agents (Node.js/TypeScript, tree-sitter, SQLite): symbol indexing across 34 languages, token-efficient retrieval (~99.9% fewer tokens than reading source) and change-impact tools (blast radius, co-change risk, refactor safety).
- Benchmarked on 87 open-source projects with 25 ground-truth queries each; in the head-to-head on 79 of them, ahead of an existing code-search MCP on 66 on Precision@1 (72 on P@3). Every release gated by an automated regression guard and blind hold-out repositories. github.com/goranocokoljic/pure-context

Toprope | Product in development, 2026

- Building Toprope, a product for monitoring and improving AI adoption in engineering organisations. tr-harness was created to develop it and is now building the companion site of this application, whose run record is public.

EARLIER EXPERIENCE

Senior Software Developer | Nil May 2020 - Mar 2021

- Built the front end (Vue 2) of a unified payment system used by Serbian public-sector organizations: ministries, municipalities and public institutions.

Founder & CTO / Software Architect / Lead Developer | Ntropy Analytics (formerly PassBy) Jun 2016 - Mar 2020

- Designed the end-to-end IoT analytics system, from custom edge devices to Kafka/Node.js services, TimescaleDB and analytics applications; processed ~900M raw rows/day across ~600 devices at peak.
- Designed staged OTA updates with canary releases, device health monitoring and rollback paths; owned architecture, product direction and hiring for a team of up to 8 developers.

Senior Full-Stack Developer / Product Owner / CMO | Factory Worldwide - VisMedic Feb 2014 - May 2016

- Built much of an end-to-end telehealth platform across HTML/SCSS/JavaScript and Python/Flask/MySQL/WebRTC, covering patient, doctor and healthcare-provider workflows.
- Shaped product direction, UI/visual identity and clinic partnerships; the platform, designed to HIPAA requirements, was used by real doctors and patients through BelMedic, MediGroup and smaller private clinics.
- Ring Serbia 2016 Winner - Best Startup; Catch Pitch Challenge 2016 Winner; Ring Balkans 2016 Runner-up (unofficial); Pioneers of the Balkans Finalist.

Creative Director, Online Media | McCann Erickson / MRM Worldwide 2007 - 2014

- Recruited to build McCann's digital department; led a creative team of up to 9 while remaining hands-on developing websites and interactive digital experiences.
- Lead presenter for client pitches with a 100% win rate across the final three years; worked with brands including Coca-Cola, General Motors, Audi, Telekom Srbija and L'Oreal.
- SEMPL 2011 Golden Award; 20+ local UEPS awards and multiple regional awards; board and awards-jury member of UEPS.

CONSULTING, IN PARALLEL WITH NTROPY

2017 - 2020 Software Architect / Consultant, FleetDirect - custom JSON-driven JavaScript UI framework plus Android fleet-tracking application.

TECHNICAL

FRONTEND JavaScript, TypeScript, Vue.js, Nuxt.js, HTML, SCSS/CSS, Vite, SSR/hybrid rendering, Core Web Vitals, SEO, headless/API-driven CMS; Astro on current personal work

BACKEND & DATA Node.js, Python/Flask, Kafka, TimescaleDB, MySQL, PostgreSQL, SQLite, WebRTC, Docker, AWS, Cloudflare

AI & DEVELOPER TOOLING Claude Code (daily), MCP (author of PureContext), Anthropic/OpenAI/Gemini APIs, agentic workflows, AI code review, RAG, eval/feedback loops, telemetry, GitHub CLI

ENGINEERING System architecture, performance, web security and code review, Playwright/E2E testing, CI/CD (GitHub Actions, TeamCity), SonarQube, Bitbucket/GitHub, mentoring

EDUCATION, TEACHING & RECOGNITION

EDUCATION Diplomirani Inženjer (Master's equivalent), IT Sciences - Faculty of Electrical Engineering, University of Belgrade

TEACHING University Lecturer, 2005 - 2017 - Web Design, Multimedia, Computer Graphics and Computer Animation across three institutions

LANGUAGES Serbian - native | English - fluent